

CosmoSort

Next Generation Spatial Cell Sorter

CosmoSort | Spatially-resolved Laser Activated Cell Sorting (SLACS)

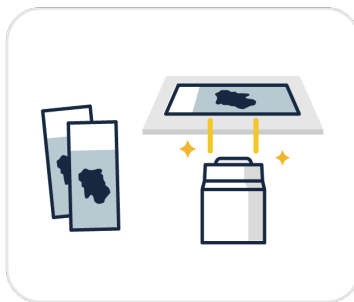


- **Dimensions**
54.5 × 58.3 × 37.8 cm
- **Weight**
Approximately 50 kg

Features

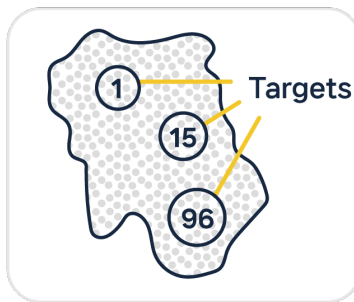
- **High precision meets high speed (1 target/second)**
CosmoSort, combining the precision of spatial sorting with the speed of high-throughput analysis, is capable of sorting cells at the remarkable rate of one target per second.
- **Innovative IR pulse laser technology**
IR pulse laser delicately isolates target cells, maintaining their natural state for accurate downstream analysis.
- **Seamless integration for diverse applications**
CosmoSort's versatility extends across various molecular assays, making it suitable for genomics, proteomics, and more.
- **Tailored flexibility for diverse research needs**
CosmoSort adapts to a wide range of research applications, from complex tissue analysis to large-scale cell sorting.

Procedure



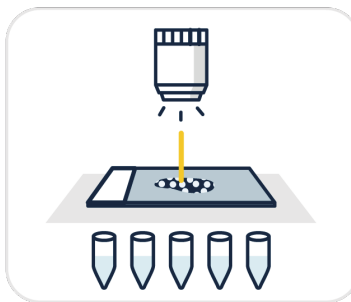
Sample preparation and imaging

Place your sample on a CosmoSlide, and apply appropriate staining techniques—such as hematoxylin and eosin (H&E), immunofluorescence (IF), or immunohistochemistry (IHC)—to enhance the visibility of specific cells or structures.



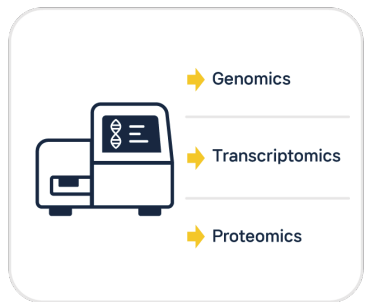
Target selection

Based on the imaging data, researchers select target regions of interest (ROIs). This selection is typically guided by specific cellular characteristics, such as the expression of certain biomarkers. Advanced software can be used to analyze the imaging data and assist in identifying the regions for sorting.



Spatial cell sorting

SLACS precisely isolates the selected cells from the rest of the sample. This is achieved through laser activation, where a focused laser beam is used to target and dislodge the selected ones without affecting adjacent cells. The isolated cells are then collected for further analysis.



Spatial omics analysis

The isolated cells then undergo full-length sequencing of preferred kind. By analyzing these profiles, researchers can gain genomic, transcriptomic, epitranscriptomic, and proteomic insights into the cellular functions, interactions, and roles of these cells in the broader biological context of the sample.